

AI in Industry, ETH Zurich, autumn 2026

The Word, the Thought, the Power, the Act

Four blocks, four voices, one order

Frontispiece: Rembrandt van Rijn, A Scholar in His Study ('Faust'), c. 1652. Etching, drypoint and burin. The Metropolitan Museum of Art, New York. CC0



Johann Wolfgang von Goethe
1749 to 1832

Faust in his study

In the study a scholar opens the Gospel of John, which begins in Greek *Ἐν ἀρχῇ ἦν ὁ λόγος*. He wants it in *mein geliebtes Deutsch*, his beloved German.

Faust I, the study. English by Bayard Taylor

Opposite: Joseph Karl Stieler, Goethe at 79, 1828. Neue Pinakothek, Munich. Wikimedia Commons, public domain



Rembrandt van Rijn, A Scholar in His Study ('Faust'), c. 1652. Etching, drypoint and burin. The Metropolitan Museum of Art, New York. CC0

Im Anfang war das Wort.
In the Beginning was the Word.

Im Anfang war der Sinn.
In the Beginning was the Thought.

Im Anfang war die Kraft.
In the Beginning was the Power.

Im Anfang war die Tat.
In the Beginning was the Act.

Goethe, Faust I, the study. English by Bayard Taylor



荀子

Xunzi
c. 310 to after 238 BC

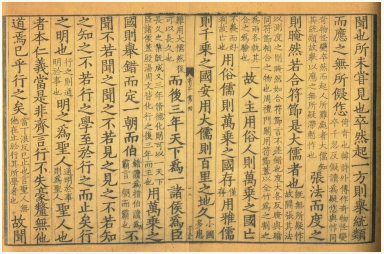
A teacher's warning

He warned that each step taken alone fails: *hearing* without *seeing*, *seeing* without *knowing*, *knowing* without *practice*.

Xunzi, chapter 8, 儒效. English by the lecturer

Opposite: Xunzi, Qing dynasty portrait, from an album of portraits of sages. Wikimedia Commons, public domain

不聞不若聞之，
聞之不若見之，
見之不若知之，
知之不若行之，
學至於行之而止矣。



Xunzi, chapter 8, with the commentary of Yang Liang. Song printed edition. National Central Library, Taiwan. Public domain

Each is better than the one before.
Learning ends in practice.

Xunzi, chapter 8. English: lecturer

Hearing: the Word.
Seeing and knowing: the Thought.
Practice: the Act.



T. S. Eliot
The Rock, 1934

Words for a pageant

Eliot wrote the words: "I cannot consider myself the author of the 'play', but only of the words which are printed here."

Title page and prefatory note, *The Rock*, 1934

Opposite: Lady Ottoline Morrell, T. S. Eliot, 1923. Photograph. Wikimedia Commons, public domain



Lady Ottoline Morrell, T.S. Eliot, 1923. Photograph. Wikimedia Commons, public domain

Knowledge of words, and ignorance
of the Word.

Where is the Life we have lost
in living?
Where is the wisdom we have lost
in knowledge?
Where is the knowledge we have lost
in information?

The Rock, 1934, part I, the opening chorus

Information: the Word.
Knowledge: the Thought.
Wisdom: the Act.



John Boyd
1927 to 1997

A fighter pilot's loop

Boyd arrived in Korea in March 1953 and flew 22 missions in the F-86 Sabre, in which he did not fire his guns or score a kill.

Wikipedia

Opposite: F-86 Sabres of the 51st Fighter Interceptor Wing over Korea, 1952. U.S. Air Force photograph

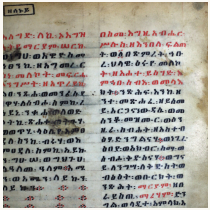


U.S. Air Force, John Boyd, probably in Korea, 1953. Photograph. U.S. Government. Public domain

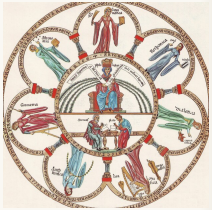
The picture of the world we carry shapes what we observe, what we decide and what we do.

Boyd, The Essence of Winning and Losing, 1995. Paraphrase

Observation: the Word.
Orientation: the Thought.
Decision and action: the Act.



The Word



The Thought



The Power



The Act

Ethiopian prayer book in Ge'ez, 15th century, Miami University Libraries. Hortus Deliciarum, c. 1180. Mira (Blue Gene/Q), Argonne National Laboratory, U.S. Department of Energy. F-86 Sabres over Korea, 1952, U.S. Air Force.

In four blocks

The course, autumn 2026

Weekend 1 *16 and 17 October* *You are here*

The Word Large language models

Ge'ez games, prompting, OPRO, MLOps

Weekend 2 *13 and 14 November*

The Thought Representation learning

The Power Large-scale computing

Recommendation systems, matrix factorization, bandits

Weekend 3 *11 and 12 December*

The Act Reinforcement learning

MDPs, Bellman equations, SARSA, REINFORCE